

One Record per Click: Serving Capacity Selects the Training Target in Delayed-Feedback Conversion Prediction

Anonymous Author¹

Anonymous Institution, Anonymous City, Country
`anonymous@example.org`

Abstract. Many production conversion-rate (CVR) pipelines write exactly one training record per click: it is emitted after a fixed wait Δ , carries the label $\mathbf{1}[\text{conversion} \leq \text{click} + \Delta]$, and is never revisited. We show that this contract reduces the choice of training target to a one-dimensional design problem — how much of the target is a realised hard label and how much a model-produced soft one — and that the served model’s capacity selects where on that axis to sit. Instrumenting the feature path of fourteen published methods over a full 720-hour replay of the Criteo Delayed Feedback Dataset (Criteo-DFD), we find that nine emit more than one record per click, including the VANILLA baseline the field normalises against, so their reported gains are not attainable under the contract. Sweeping target hardness over the admissible remainder against a five-rung serving-capacity ladder (26 K–9.6 M parameters) yields two ordered crossovers. At the smallest rung a purely soft target that uses no realised label at all is the best arm, beating even the non-causal Oracle by 0.0029 area under the ROC curve (AUC); by 46 K the hybrid targets of prior work overtake it; and beyond 10^5 parameters the Oracle’s true label is best, with the soft target now costing 0.0084. We further find that a published baseline’s prescribed auxiliary learning rate costs it 0.0022 AUC, roughly seven times the gap between the methods it is used to compare.

Keywords: Conversion rate prediction · Delayed feedback · Knowledge distillation · Online advertising.

1 Introduction

Conversion-rate (CVR) prediction decides what an advertising system shows and what it charges, and it is trained continuously on a stream. Its central difficulty is that the label arrives late: a click is observed immediately, but the conversion it may cause can follow minutes, days or weeks afterwards [1], so a model trained on the labels observed so far is trained on a censored, downward-biased target. The literature treats this as an estimation problem, and a decade of work has produced importance weighting [14, 13], label correction [2, 11], duplicated replay streams [6, 5, 3], multi-window ensembles [7, 9, 15] and influence-function

retrofits [4]. Almost all share an unstated systems assumption: that a click may be written into the training stream more than once — re-emitted when its conversion lands, replayed when its window closes, or relabelled at several horizons.

Many production pipelines cannot do this. A common contract is:

A click produces *exactly one* training example. It is emitted after a fixed wait Δ , carries the label $\mathbf{1}[\text{conv_ts} \leq \text{click_ts} + \Delta]$, and is never revisited. If the conversion has not arrived by then, the example is negative permanently.

The reasons are mundane and therefore durable: amending a materialised example later means mutable training data, duplicate-key handling, and a reproducibility story that breaks on every backfill, whereas freezing the label at Δ keeps the dataset append-only and the pipeline replayable. The contract is not our invention. It is the literal reading of the field’s own reference baseline, which “trains each click *once* at base observation age *o*” [8]. What we show is that the implementations do not honour it — including, as Sect. 5.2 establishes, that baseline’s own.

One record, one question. The contract is restrictive enough to be clarifying. If a click yields a single record carrying a single target, the only remaining freedom is *what that target says*. At one extreme it is a realised outcome: a hard 0/1 draw, unbiased but high-variance, and correct only for conversions fast enough to have landed. At the other it is a model’s estimate: a soft probability, lower-variance and defined for every record, but capped by the estimator that produced it. Every admissible method is a point between these two. FSIW [14] reweights the hard label, ULC [11] blends it with a learned correction, and TWICE [8] routes the residual to a separate head. We call this quantity the *hardness* of the target, and we show that the best choice along it is not a property of the dataset but of the *served model’s capacity* — the axis a latency-bound ranker is actually constrained on, and the one the delayed-feedback literature holds fixed.

Contributions.

1. **A measurement, not an assertion.** We instrument the feature path of fourteen methods and count training records per click over a full replay (Sect. 5.2). Nine exceed one. Two measurement traps — traffic variation and window length — both *understate* violations; we control for both.
2. **A reformulation.** Under the contract the admissible targets form a single hardness axis spanned by two label pipelines (Sect. 3), and a purely soft target satisfies the frozen-label obligation *vacuously*, its record carrying no label to freeze (Proposition 1).
3. **The main result.** Optimal hardness increases with serving capacity, through two ordered and significant crossovers (Sect. 5.4): the soft target cedes to the hybrid between 26 and 46 K parameters, and the hybrid cedes to the Oracle’s true label near 10^5 . Below that second crossing the non-causal Oracle is *not* an upper bound.

4. **A reproducibility finding.** Baseline auxiliary learning rates do not transfer: one baseline’s prescribed rate costs it 0.0022 AUC while the same change *harms* another (Sect. 5.3), so the admissible ordering cannot be read without per-method tuning.

2 Related Work

We organise prior work by whether it can be expressed as one record per click, because that is what the constraint decides. Sect. 5.2 gives measured counts; here we give the mechanism.

Methods that need more than one record. The fake-negative family writes the click as a negative and re-emits it as a positive when the conversion arrives [6]; ES-DFM samples an elapsed-time cut-off and duplicates the delayed positive [13], DEFER replays real negatives once their window closes [5], DDFM runs both streams together [3], and DEFUSE adds a label correction on top of the duplicated stream [2]. Multi-window methods relabel the same click at several horizons: FTP trains K maturity-gated tasks and a policy over them [7], MISS duplicates each delayed positive into every head’s stream [9], and PI interpolates two fixed-window predictors [15]. IF-DFM applies an influence-function update when a label reverses [4]. In each case the extra visit *is* the mechanism, not an implementation detail that could be removed.

Single-record methods, and where they sit on the axis. A smaller group reweights or relabels the one fresh record, and these are what the constraint leaves standing. FSIW keeps the realised label and importance-weights the loss [14]; ULC adds a learned correction to it, giving a target that is part observation and part estimate [11]; TWICE keeps its CVR tower on a single fresh record and routes conversion-clock events to a separate delay head [8]. All three retain the record’s own realised label; none removes it. The end of the axis at which the target is a model estimate alone is, to our knowledge, unmeasured in this literature, and Sect. 5.4 shows it is the best admissible choice at small serving capacity.

Distillation with privileged information. Training-time-only signals have been distilled into serving models before, notably privileged features at Taobao [12], and soft labels carry lower target variance than a hard Bernoulli label [10] — the mechanism we invoke in Sect. 5.4. The privilege we exploit is neither a feature nor scale but *time*: the estimator producing the soft target is allowed to wait.

3 Problem Formulation

3.1 Notation and two clocks

Table 1 fixes notation. Following the long-horizon formulation of [8], clicks are observed on a *click clock* and conversions arrive on a separate *conversion clock*: a business attribution window v fixes the served target $Y^{(v)}$, and training must predict $Y^{(v)}$ from a record written before $Y^{(v)}$ is known.

Table 1. Notation. Times are absolute seconds; $o \ll v$ throughout

symbol	meaning
x_i, C_i	features and click time of click i
$V_i, D_i = V_i - C_i$	conversion time and delay ($D_i = \infty$ if no conversion)
v	attribution window; the business-defined horizon of the served target
o	base observation age: the shortest wait at which a record may be emitted
Δ	the pipeline’s actual wait before emitting the record
$Y_i^{(v)} = \mathbf{1}[D_i \leq v]$	the served target, knowable only at $C_i + v$
$Y_i^{(o)} = \mathbf{1}[D_i \leq o]$	the label observable at emission when $\Delta = o$
$p_v(x) = P(Y^{(v)}=1 \mid x)$	the estimand; a model of it is written $\hat{p}_v$
$w(x) = P(Y^{(v)}=1 \mid Y^{(o)}=0, x)$	the residual correction shared by FSIW, ULC, ES-DFM
$\hat{y}_i$	the target actually written on click i ’s record

3.2 The single-visit constraint

Let Δ be the pipeline’s wait. The contract of Sect. 1 imposes two obligations:

- R1 (single visit)** click i contributes exactly one training record, ever;
R2 (frozen label) that record’s label is $\mathbf{1}[D_i \leq \Delta]$ for the Δ at which it was emitted; no later-arriving conversion may add or change supervision.

A method may fail either independently: duplication fails R1, and reading $Y^{(v)}$ on a record released before $C_i + v$ fails R2.

3.3 Two label pipelines

Because a record is written once and Δ is the only free quantity, the constraint admits exactly two useful pipelines over the same clicks:

$$\mathcal{D}_{\text{fresh}} = \{(x_i, Y_i^{(o)})\}, \quad \Delta = o \ll v \quad \text{timely, biased low,} \quad (1)$$

$$\mathcal{D}_{\text{mat}} = \{(x_i, Y_i^{(v)})\}, \quad \Delta = v \quad \text{unbiased, stale by } v. \quad (2)$$

For $\mathcal{D}_{\text{fresh}}$, $\mathbb{E}[Y^{(o)} \mid x] = p_v(x) F(o \mid x) < p_v(x)$, so it is biased low by exactly the fraction of converters not yet observed. On Criteo-DFD at $o = 1$ h that fraction is 0.77, 0.57 and 0.47 at $v = 1, 7$ and 30 d: censoring deepens as the window lengthens, yet even at $v = 30$ d the fresh label still carries almost half the target’s positives, which makes this corpus a conservative setting for any method that discards it. $\mathcal{D}_{\text{mat}}$ carries the true target but its most recent row is v old.

R1 forbids training the served model on both. What remains is to write one record at $\Delta = o$ and use $\mathcal{D}_{\text{mat}}$ to shape *what its target says*.

3.4 The target-hardness axis

Every admissible target is a function of the record's own realised label $Y_i^{(o)}$ and of quantities estimated on $\mathcal{D}_{\text{mat}}$. Writing the general single-record form

$$\tilde{y}_i = Y_i^{(o)} \alpha(x_i) + (1 - Y_i^{(o)}) \beta(x_i), \quad (3)$$

with α, β fitted on $\mathcal{D}_{\text{mat}}$ and frozen at emission, gives a one-dimensional family ordered by how much the target depends on the realised label. We name three points on it, and they are the arms we compare:

hard the target *is* a realised label: $\alpha=1, \beta=0$ recovers $\tilde{y} = Y^{(o)}$ (FRESH); the non-causal reference $\tilde{y} = Y^{(v)}$ (ORACLE) lies outside the family, since $Y^{(v)}$ is not knowable at $\Delta = o$;

hybrid realised label plus a learned correction: $\alpha=1, \beta=w$ gives $\tilde{y} = Y^{(o)} + (1 - Y^{(o)})w(x)$, which is exactly ULC [11];

soft a model estimate alone: $\alpha=\beta=\hat{p}_v$ gives $\tilde{y} = \hat{p}_v(x)$, which uses no realised label at all.

FSIW [14] sits at the hard end by a different route, leaving $\tilde{y} = Y^{(o)}$ and reweighting the loss instead of the target.

Proposition 1. *Let $\tilde{y}_i$ follow (3) with α, β fitted on $\mathcal{D}_{\text{mat}}$ and frozen at emission time $C_i + o$. Then (i) $\tilde{y}_i$ is computable at $\Delta = o$ and the record satisfies R1 and R2; and (ii) if $\alpha \equiv \beta$, the record satisfies R2 vacuously: its target is a function of x_i alone, so it carries no realised label that a later-arriving conversion could revise.*

Proof. (i) The record is written once, at $\Delta = o$, so R1 holds. Its target is a deterministic function of $(x_i, Y_i^{(o)})$ and of parameters frozen at emission; $Y_i^{(o)}$ is knowable at $C_i + o$ and is never revised, so R2 holds. (ii) If $\alpha \equiv \beta$ then (3) collapses to $\tilde{y}_i = \alpha(x_i)$, which does not depend on $Y_i^{(o)}$. R2 constrains the realised label carried by the record, and there is none. $\square$

Proposition 1(ii) is the structural reason to care about the soft end beyond accuracy: a record with no realised label needs no label join on the timely path. We return to this in Sect. 6.

4 Instantiating the Axis

Realising the hybrid and soft points of Sect. 3.4 requires an estimator of w and of p_v that is itself admissible. We fit one model on the matured pipeline and read both from it, so that a comparison across the axis changes the target and nothing else.

The estimator trains only on clicks whose window has closed, $C_i + v \leq T_n$ at cutoff T_n , under the true label $Y^{(v)}$. It is causal by construction — every

training row’s outcome was knowable at T_n — and is itself R1/R2-compliant at $\Delta = v$. It carries two read-outs on one trunk:

$$\hat{p}_v(x) = P(Y^{(v)} = 1 \mid x), \quad w^T(x) = P(Y^{(v)} = 1 \mid Y^{(o)} = 0, x). \quad (4)$$

The second is the estimand the field shares — it is ULC’s w , ES-DFM’s delayed-positive rate and DDFM’s z_1 . Substituting (4) into (3) gives the two admissible arms we sweep: the hybrid $\tilde{y} = Y^{(o)} + (1 - Y^{(o)}) w^T(x)$, and the soft $\tilde{y} = \hat{p}_v(x)$. Because the estimator is training-only it is unbound by the serving budget; that asymmetry is what buys the right to wait.

Two implementation points matter and are easy to get wrong; both cost us a run before they were fixed. First, the estimator’s learning rate must be *decoupled* from the served model’s: sharing it makes a per-arm sweep tune two models with one scalar, which cost 0.0065 AUC when it happened to FSIW’s auxiliary in our harness. Second, both read-outs must be withheld until the estimator has seen a minimum number of matured rows — the hybrid falling back to $\tilde{y} = Y^{(o)}$ and the soft arm likewise — because a zero-initialised head asserts 0.5 on every row, and a constant target yields a loss of $\ln 2$ at every learning rate. Without the guard the soft arm’s rate is selected on a degenerate objective; Sect. 7 records the one window where this bites even with the guard in place.

5 Experiments

5.1 Setup

Dataset and protocol. Criteo-DFD [1], 60 days of clicks with conversion timestamps. We follow the streaming replay of [8]: pre-train on days $[0, 30)$, then stream $[30, 60)$ with hourly cut-offs; at each cut-off a model consumes the records released in that hour and then predicts every click of the next hour exactly once, scored retrospectively against $Y^{(v)}$. Base observation age $o = 1$ h. We report $v = 7$ d in the main text and $v = 1$ d as a second window.

Evaluation metrics. Area under the ROC curve (AUC) is the headline metric, with area under the precision–recall curve (PR-AUC) and log-loss (LL) reported at the capacity-matched rung. All numbers are means over five seeds, and every comparison is a paired t -test over matched seeds ($*p < 0.05$, $**p < 0.01$).

Compared methods. Table 3 names every arm and its point on the axis. REWEIGHT is FSIW [14], CORRECT is ULC [11], WAIT trains on $\mathcal{D}_{\text{mat}}$ only, and DISTIL and SOFT use the estimator of Sect. 4. VANILLA, ES-DFM [13], DEFUSE [2] and DDFM [3] are shown for reference but are inadmissible (Sect. 5.2). The non-causal Oracle labels the fresh record with $Y^{(v)}$.

Implementation details. Minibatch 8192, AdamW, and a learning rate selected *per arm and per capacity* from $\{10^{-6}, 2.5 \times 10^{-6}, 10^{-5}, 5 \times 10^{-5}, 2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$ on a held-out chronological slice of the pre-training period. The served model is swept over five rungs S0–S4 varying the categorical hash-table size ($36 \rightarrow 131,072$ rows, $26 \text{ K} \rightarrow 9.6 \text{ M}$ parameters) at a fixed embedding dimension of 8, so the ladder is a pure serving-capacity axis. The matured-data estimator is held at 9.5 M

Table 2. Training records per click, measured over a full 720-hour replay at $v = 7$ d. Counts are per *distinct click touched*, so traffic variation cannot inflate them, and the window covers every method’s longest revisit lag. [†]Twice’s served tower is single-visit; its delay head reads the conversion clock

method	records/click	max	R1	R2
<i>admissible</i>				
Fresh	1.00	1	✓	✓
Wait / teacher	1.00	1	✓	✓
Reweight	1.00	1	✓	×
Correct	1.00	1	✓	×
Twice	1.00	1	✓	✓ [†]
<i>inadmissible</i>				
Vanilla	1.08	2	×	×
DFM	1.34	8	×	×
ES-DFM	1.08	2	×	×
DEFUSE	1.08	2	×	×
DDFM	1.89	2	×	×
PI	2.00	2	×	×
MISS	2.05	3	×	×
IF-DFM	2.90	99	×	×
FTP	3.00	3	×	×

parameters throughout. At S4 the served model and the estimator are within 1.3% of the same parameter count, which is the cleanest available control for capacity transfer. Auxiliary learning rates are swept per method for the reason established in Sect. 5.3.

5.2 RQ1: which methods can be run at all?

Table 2 reports the measurement. We instrument the feature path of every arm and count training records over a full 720-hour replay, charging only gradient-carrying forwards so that merely *scoring* a historical click is not counted. Two traps had to be controlled, and ignoring either understates violations.¹

Nine of fourteen methods exceed one record per click. Two counts are exact integers and confirm the instrument: PI emits 2.00, one per fixed-window head, and FTP 3.00, one per task plus its prophet. IF-DFM touches one click up to 99 times. Only FSIW, ULC and TWICE keep the served model single-visit.

The field’s anchor is inadmissible. VANILLA — “each click trained once at base observation age o ” — measures 1.08 records per click, because matching any

¹ *Traffic*: per-hour ratios conflate duplication with diurnal variation, since a maturity-gated stream reads clicks from v ago when the rate differed; PI scores a spurious 12.2 that way. We divide by *distinct clicks touched*. *Window*: over 48 h PI measures 1.00, its two visits being 10.5 days apart. We run the full stream and score only clicks whose entire revisit horizon fits inside it.

published VANILLA row requires also emitting the late conversion as a positive.² Relative improvement reported against it is therefore normalised by a baseline a single-visit pipeline cannot run.

On auxiliaries. FSIW, ULC and TWICE emit one record for the served model but fit a *separate* model on matured data, so they satisfy R1 and sit at the boundary of R2. We read the constraint as governing the served model’s training pipeline, under which they comply; the alternative reading — that no later-arriving conversion may inform anything — leaves only WAIT standing and declares the problem unsolvable rather than selecting a method. We adopt the former and say so, because the choice determines which methods are admissible. Every arm we compare sits at exactly this level and no lower.

5.3 RQ2: is the admissible ordering readable?

Before comparing targets we must establish that the comparison is about targets. Each single-record baseline prescribes a fixed learning rate for its auxiliary, and these are not interchangeable. Holding everything else fixed and moving CORRECT’s auxiliary from its prescribed 10^{-3} to 5×10^{-4} raises it by $+0.0005^*$, $+0.0004^{**}$, $+0.0002^*$, $+0.0002^{**}$ and $+0.0022^{**}$ across S0–S4. The same change applied to REWEIGHT *harms* it, by -0.0005^{**} , -0.0004^{**} and -0.0036^{**} at S0, S2 and S3.

Two consequences follow. First, the effect is large relative to what it is used to measure: at S4 the tuning alone moves CORRECT by 0.0022, roughly seven times the 0.0003 that separates it from DISTIL once both are tuned. A comparison at prescribed rates would have reported a 0.0025 method gap that is 88% tuning. Second, because the same change helps one baseline and hurts another, the bias is not systematic and cannot be removed by a global correction. We therefore sweep auxiliary rates per method and report the tuned value for every arm; all numbers below are tuned. We flag this rather than bury it because it bounds how finely published orderings in this literature can be read.

5.4 RQ3: optimal target hardness tracks serving capacity

Table 3 and Fig. 1 are the main result, and the table is read down the columns rather than across the rows: the best admissible target changes with capacity.

First crossover: soft cedes to hybrid between 26 and 46 K parameters. At S0 the soft target is the best admissible arm, ahead of the hybrid by $+0.0012^*$. By S1 the ordering has reversed, and it stays reversed and widens monotonically: -0.0038^{**} , -0.0043^{**} , -0.0072^{**} and -0.0084^{**} at S1–S4. The soft arm’s own ladder is nearly flat ($0.8285 \rightarrow 0.8352$ across the five rungs) where the hybrid’s climbs ($0.8273 \rightarrow 0.8437$), which is the mechanism: a purely soft target is a *function of x* , so two records with identical features receive identical supervision,

² The fresh record alone predicts 0.107 against a 0.227 base rate and cannot reproduce any published VANILLA log-loss; the reading that does is fresh label plus a positive for conversions arriving after o .

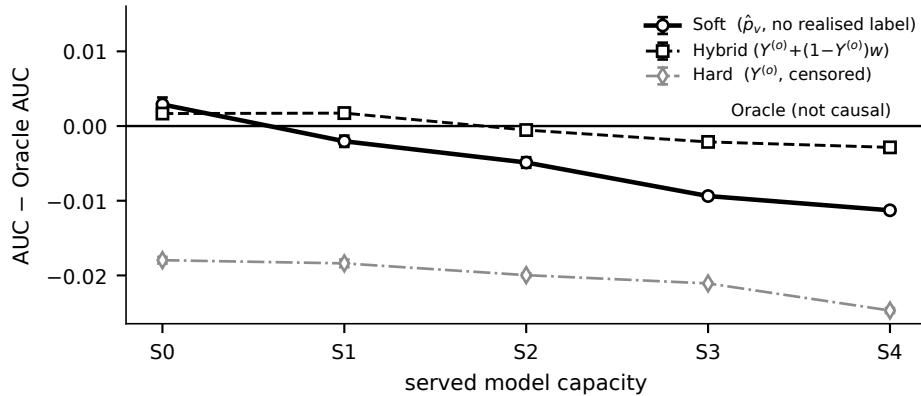


Fig. 1. Both crossovers, plotted as a margin against the non-causal Oracle so that the Oracle is the zero line; bars are 95% intervals over five paired seeds. The soft and hybrid targets cross *each other* between S0 and S1, and both cross *zero* between S1 and S2: below roughly 10^5 served parameters an estimated target beats the true one

and a larger model can only fit that function more exactly. The realised label carries per-example information that no function of x reproduces. A small model cannot exploit that information and is hurt by its Bernoulli variance [10]; a large one can, and discarding it becomes pure loss.

Second crossover: the Oracle is not an upper bound below 10^5 parameters. The hybrid beats the non-causal Oracle at S0 and S1, by $+0.0017^{**}$ at both, and loses from S2 on (-0.0005^* , -0.0021^{**} , -0.0029^{**}). The soft target beats it more decisively still at S0, by $+0.0029^{**}$. Below roughly 10^5 parameters a corrected or estimated target is therefore *better supervision than the true label*, for the same variance reason. This matters for how the literature evaluates: an assessment conducted only at large capacity reports a delayed-feedback penalty that a deployable model does not fully pay.

Within the hybrid class, the source of the correction barely matters. DISTIL and CORRECT differ by -0.0007 to $+0.0004$ across the ladder, reaching $p < 0.05$ only at S3 and S4 and never exceeding 0.0003 there. This is a parity result, and it is informative: the two correctors hold the same parameter count to within 0.13% (9,444,801 against 9,456,897) and are fit on the same schedule over the same matured stream, so the comparison is capacity-matched at the corrector as well as, at S4, at the served model. Having controlled for capacity and for tuning, what remains is small. The practical reading is that the choice of hybrid estimator is not where the value lies; the choice of *hardness class* is.

Compliance is nearly free. The strongest inadmissible method, DDFM, reaches 0.8436 at S4 against the best admissible 0.8437, and trails from S1 upward. Giving up duplication costs essentially nothing on this corpus, which is the result that makes the constraint livable rather than merely binding.

Table 3. AUC against serving capacity, $v = 7$ d, five seeds, arms grouped by their point on the hardness axis of Sect. 3.4. Bold marks the best *admissible* arm per column; the Oracle is not causal and the inadmissible block cannot be run at all, so neither competes for it. The bold entry moves from the soft target at S0 to the hybrid from S1 on

training target	S0	S1	S2	S3	S4
served parameters	26K	46K	115K	645K	9.6M
<i>hard: target is a realised label</i>					
FRESH ($\tilde{y} = Y^{(o)}$)	0.8077	0.8127	0.8182	0.8227	0.8218
REWEIGHT (loss-weighted $Y^{(o)}$)	0.8218	0.8269	0.8334	0.8388	0.8396
WAIT ($\tilde{y} = Y^{(v)}$, stale by v)	0.8105	0.8156	0.8212	0.8286	0.8321
<i>hybrid: realised label + learned correction</i>					
TWICE	0.8120	0.8217	0.8260	0.8305	0.8327
CORRECT	0.8281	0.8325	0.8374	0.8414	0.8434
DISTIL	0.8273	0.8328	0.8377	0.8416	0.8437
<i>soft: model estimate alone, no realised label</i>					
SOFT ($\tilde{y} = \hat{p}_v$)	0.8285	0.8291	0.8333	0.8344	0.8352
<i>inadmissible: > 1 record per click</i>					
VANILLA	0.8222	0.8280	0.8347	0.8403	0.8412
ES-DFM	0.8166	0.8288	0.8357	0.8383	0.8421
DEFUSE	0.8171	0.8293	0.8362	0.8387	0.8423
DDFM	0.8275	0.8314	0.8367	0.8412	0.8436
Oracle (<i>not causal</i>)	0.8256	0.8311	0.8382	0.8438	0.8465

Second window. At $v = 1$ d the ordering of the admissible field is unchanged and the Oracle inversion is stronger: at S0 the hybrid reaches 0.8352 and CORRECT 0.8334 against an Oracle of 0.8323.

5.5 RQ4: which part of the estimator matters?

Table 4 varies the matured-data estimator while holding the served model fixed. Removing its output entirely returns the hybrid to FRESH, which is the whole effect. Restricting the estimator to train only the head the served model consumes changes nothing, so carrying the second read-out is free — which is what makes a single estimator able to supply both the hybrid and the soft arm. Enlarging it from 9.5 M to 57 M or 60 M parameters does not help and slightly hurts. Criteo-DFD’s nine categorical fields hold 135,151 distinct values in total against a table provisioned at 131,072 rows *per field*, so an estimator at full resolution has already absorbed every input the data can distinguish; the saturation point and the estimator’s optimum coincide. We therefore read the estimator’s size as saturated on this corpus rather than as a tuned choice.

Table 4. Estimator design at a fixed served model (S2). Δ is against the default

variant	AUC	Δ vs default
DISTIL, teacher 9.5 M (default)	0.8377	–
teacher 57 M	0.8360	–0.0016**
teacher 60 M	0.8361	–0.0016**
teacher trains only w	0.8376	–0.0000
teacher label unused (NoKD)	0.8182	–0.0194**

6 Discussion: why the soft end is worth its cost

Table 3 makes the soft target look like a narrow win: best at one rung, clearly worse at four. Proposition 1(ii) is the reason to care about it anyway. A target that does not read the record’s realised label leaves that record with no label at all — only features and an estimator’s read-out — so R2 is satisfied with nothing to check.

This matters when one served model must predict several conversion events with different attribution windows, the normal situation in advertising: an install, a purchase and a subscription renewal do not share a horizon. Under R1 they share a single record. A hard or hybrid target requires that record to carry a realised label *per objective*; all of them can be frozen at a common o , but their informativeness differs by orders of magnitude — within this one dataset the fresh label carries 77% of the target’s positives at $v = 1$ d and 47% at $v = 30$ d (Sect. 3) — so a single shared observation age is well matched to short-window objectives and nearly vestigial for long-window ones, while still binding every corrector to the $P(Y^{(v)}=1 \mid Y^{(o)}=0)$ conditioning. A soft target removes both dependencies: one estimator per objective, each waiting its own v offline, and a served record carrying features only.

We are careful about what this licenses. Our measurements establish that the soft target is competitive only below roughly 10^5 served parameters — the regime a latency-bound ranker occupies, but a real restriction — and for one objective, not several. The multi-objective argument is a consequence of Proposition 1, not an experimental result of this paper; what the paper contributes to it is the crossover location. The cost is one additional training-time model, with serving cost unchanged, which is the trade the hybrid methods already make.

7 Limitations

One public corpus with a single conversion event, so the multi-objective consolidation of Sect. 6 is motivation rather than experiment. The constraint is one production contract among several: systems that do support replay may use the methods we rule out, and for them our audit is a cost accounting rather than a prohibition. Whether R2 binds auxiliary models is a reading we adopt explicitly (Sect. 5.2), not a fact about the constraint. The ladder varies table rows at fixed

embedding width; other capacity axes may differ. We report $v = 1$ and 7 d but not 30 d: on a 60-day corpus a 30-day window leaves the matured-data estimator with no rows at all during the model-selection slice, so the soft arm’s learning rate is chosen against a constant target and the resulting comparison measures the corpus’s span-to-window ratio rather than the effect of a long window. Finally, all results are offline; we make no online claim.

8 Conclusion

Delayed-feedback methods are usually compared on accuracy at one model size. We added a systems constraint that many production pipelines already impose — one training record per click, label frozen at a fixed wait — and found that nine of fourteen published methods cannot satisfy it, including the baseline the field normalises against. The constraint is clarifying rather than merely restrictive: with one record carrying one target, the remaining design freedom collapses to how much of that target is a realised label. Along that axis the best choice is set by the served model’s capacity, through two crossovers we locate and test. Below roughly 10^5 parameters a corrected or wholly estimated target beats the true label, and at the smallest size a target that reads no label at all is best of any admissible arm; above that the ordering reverses. Which hybrid estimator supplies the correction turns out to matter far less — 0.0003 AUC — than the tuning of the baselines it is compared against, which moves 0.0022. Evaluating at a single capacity, as this literature does, hides all of this.

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